

Assignment 8

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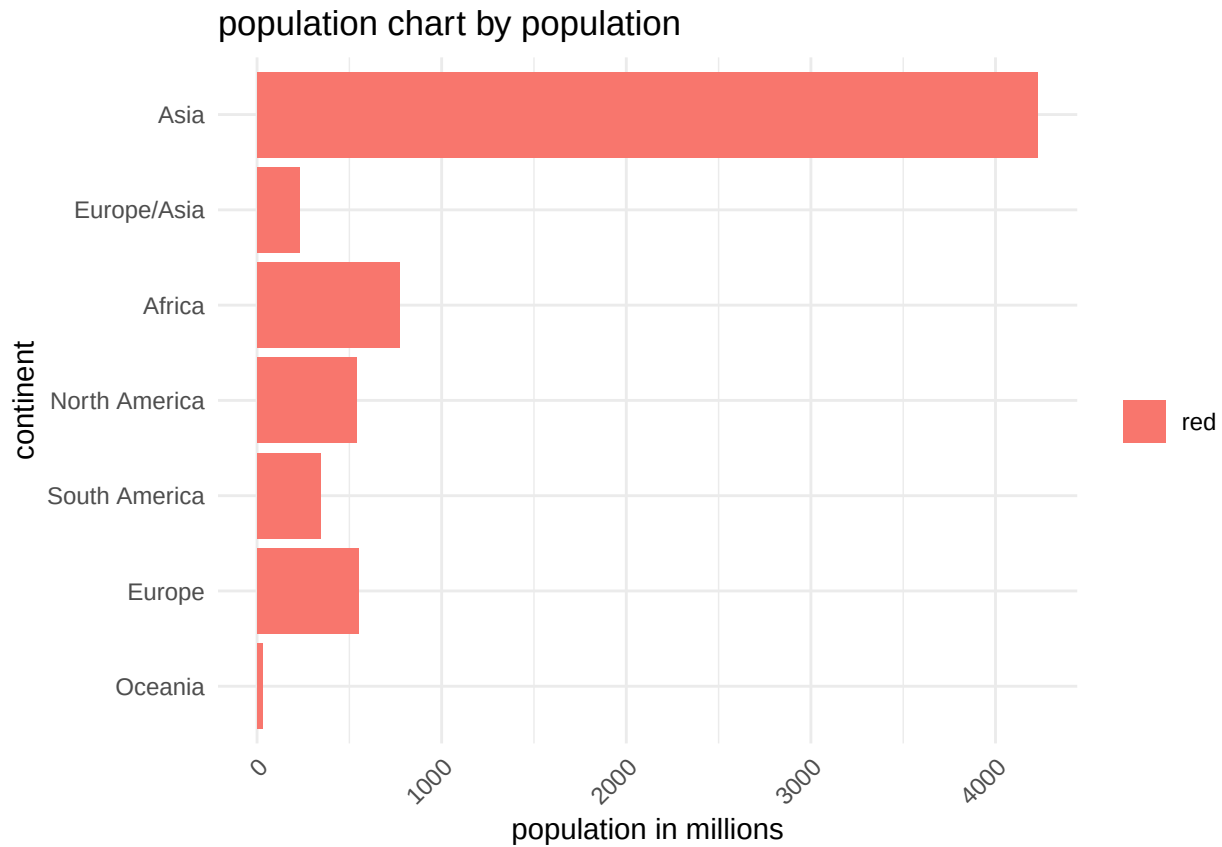
2025-11-30

```
install.packages("readxl")
install.packages("ggplot2")
install.packages("dplyr")
install.packages("tidyverse")
install.packages("treemapify")
library(readxl)
library(ggplot2)
library(dplyr)
library(tidyverse)
library(treemapify)
install.packages("devtools")
library(devtools)
devtools::install_github("htastan/TRmaps")
library(TRmaps)
install.packages("sf")
library(sf)
```

```
data("tr_nuts3")
```

```
poptr<- read_excel("2024TR.xlsx")
popglbl<- read_excel("popglbl.xlsx")
poptr2<- read_excel("tr.data.xlsx")
```

```
ggplot(popglbl, aes(x =reorder(continent, pop.in.millions) , y = pop.in.millions ,fill = "red")) +
  geom_col() +
  coord_flip()+
  labs(x
    = "continent",
    y
    = "population in millions",
    title = "population chart by population",
    fill = "")+
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45,hjust = 1))
```



A bar plot of populations by continent is provided. Asia is the most populous and densely populated continent. Oceania has the least population. There is close proximity between the other continents.

```
veri <- data.frame(
  Yas_Grubu = c("0-4", "5-9", "10-14", "15-19", "20-24", "25-29", "30-34", "35-39",
    "40-44", "45-49", "50-54", "55-59", "60-64", "65-69", "70-74",
    "75-79", "80-84", "85+"),
  Erkek = c(2606763, 3274360, 3314979, 3298654, 3235130, 3336548, 3182620, 3174570,
    3282536, 2979015, 2726368, 2301684, 2115272, 1608416, 1124799,
    716554, 374066, 243536),
  Kadin = c(2475623, 3110725, 3144115, 3129876, 3099499, 3218069, 3105980, 3125998,
    3245656, 2933900, 2739014, 2277880, 2115568, 1725547, 1316670,
    964063, 568407, 470240))

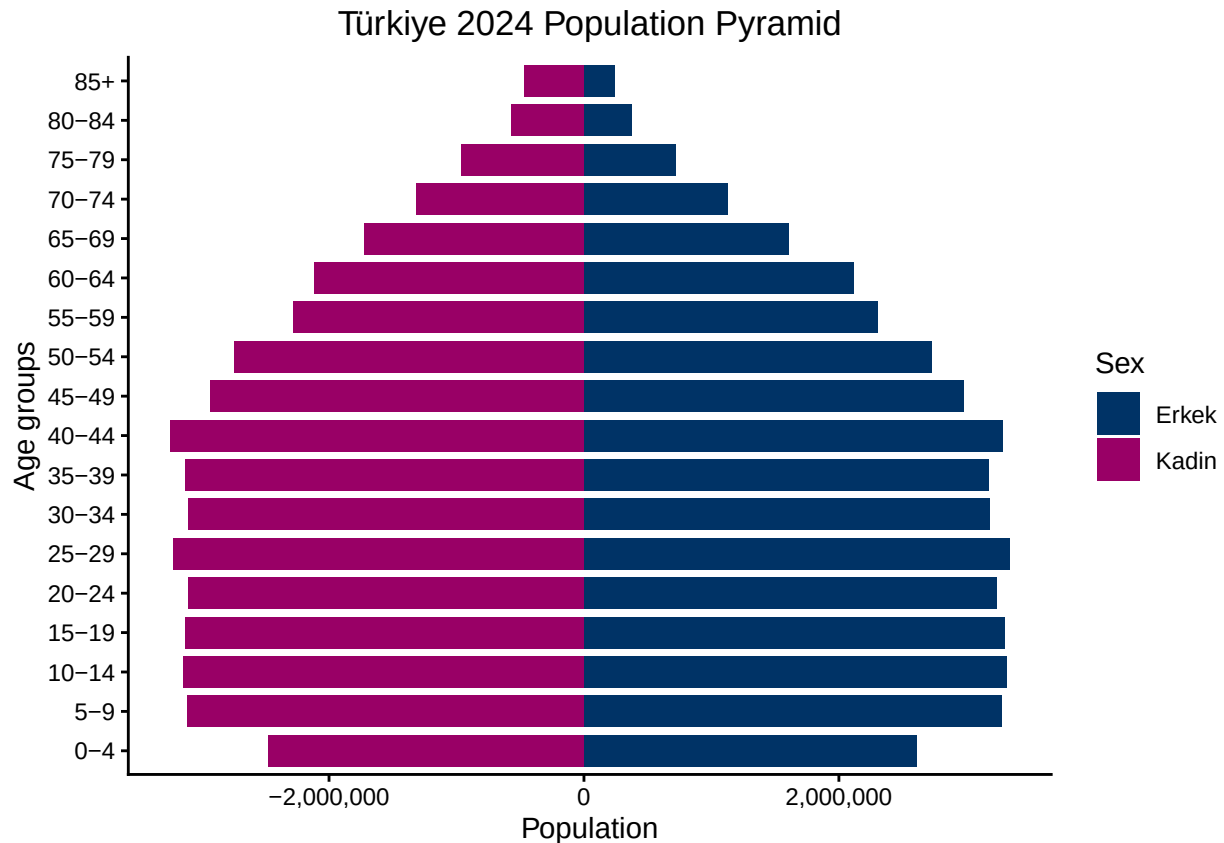
veri <- veri %>%
  mutate(Kadin = -Kadin)

veri_long <- veri %>%
  tidyr::pivot_longer(cols = c("Erkek", "Kadin"),
    names_to = "Cinsiyet",
    values_to = "Nufus")

veri_long$Yas_Grubu <- factor(veri_long$Yas_Grubu,
  levels = veri$Yas_Grubu)

ggplot(veri_long, aes(x = Yas_Grubu, y = Nufus, fill = Cinsiyet)) +
  geom_bar(stat = "identity", width = 0.8) +
```

```
coord_flip() +
scale_y_continuous(labels = scales::comma) +
labs(title = "Türkiye 2024 Population Pyramid",
      x = "Age groups",
      y = "Population",
      fill = "Sex") +
theme_classic() +
theme(plot.title = element_text(hjust = 0.5)) +
scale_fill_manual(values = c("Erkek" = "#003366", "Kadin" = "#990066"))
```



```
tr.data<- left_join(tr_nuts3, poptr2 , by = c("name_tr" = "city"))
```

```
ggplot(tr.data) +
  geom_sf(aes(fill = grwt_rate)) +
  geom_sf_text(aes(label=name_tr),size=1.5)+
  theme_void()+
  theme( plot.title= element_text(
    color="#0F2D38" , face="bold" ),
    legend.title= element_text(
    color="#0F2D38"
  ))+
  scale_fill_gradient(low = "white",high="#2C82A3")+
  labs(title="Türkiye's 2024 Population Growth Rate",
       fill="Growth Rate (%)")
```

```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not
## give correct results for longitude/latitude data
```

Türkiye's 2024 Population Growth Rate



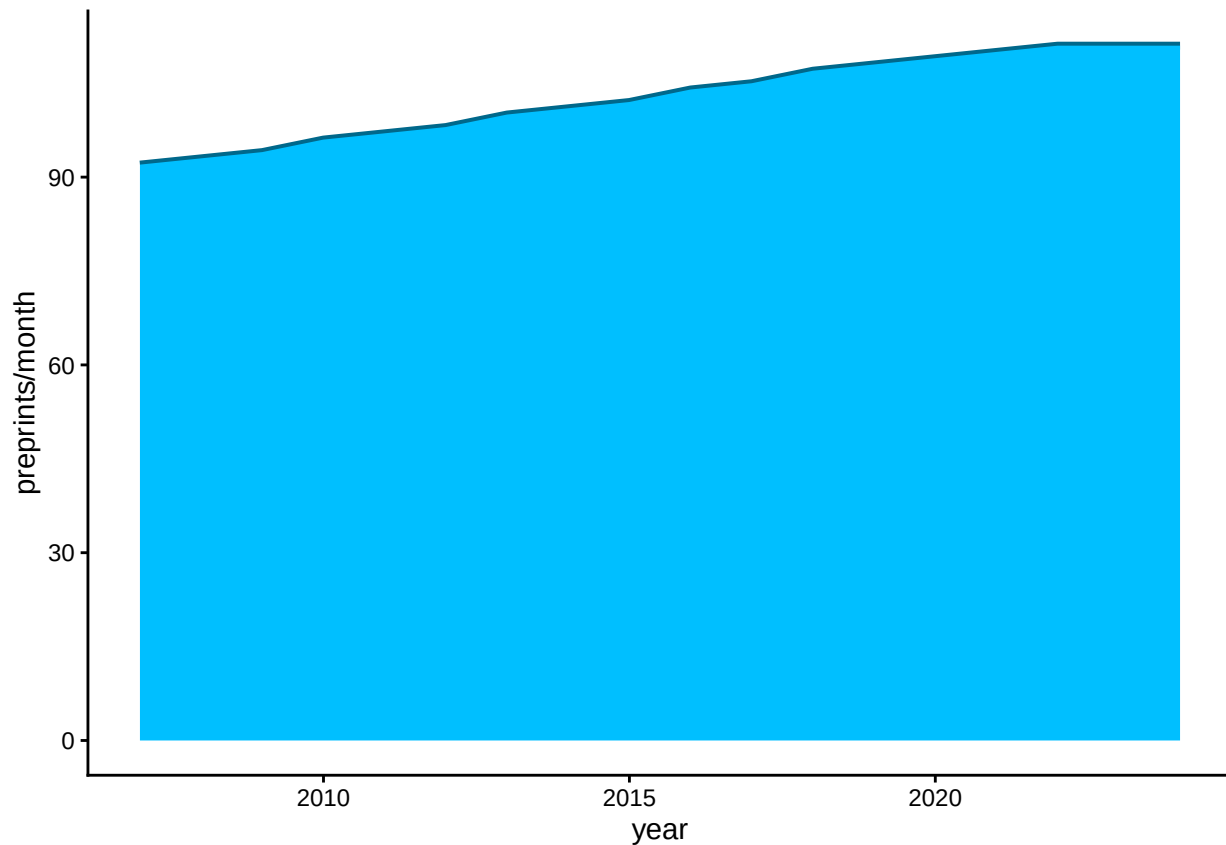
Türkiye's growth rate is given by province. The color intensity tends to darker in the eastern and western regions, while it remains lighter in the Black Sea and central Anatolia regions.

```
ggplot(poptr2, aes(x = year2, y = pop_density)) +  
  geom_line(color = "deepskyblue4", size = 1.4) +  
  theme_classic() +  
  geom_area(fill = "deepskyblue") +  
  labs(x = "year",  
       y = "preprints/month")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.  
## i Please use `linewidth` instead.  
## This warning is displayed once every 8 hours.  
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was  
## generated.
```

```
## Warning: Removed 63 rows containing non-finite outside the scale range  
## (`stat_align()`).
```

```
## Warning: Removed 63 rows containing missing values or values outside the scale range  
## (`geom_line()`).
```



```
install.packages("treemapify")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```
library(treemapify)
```

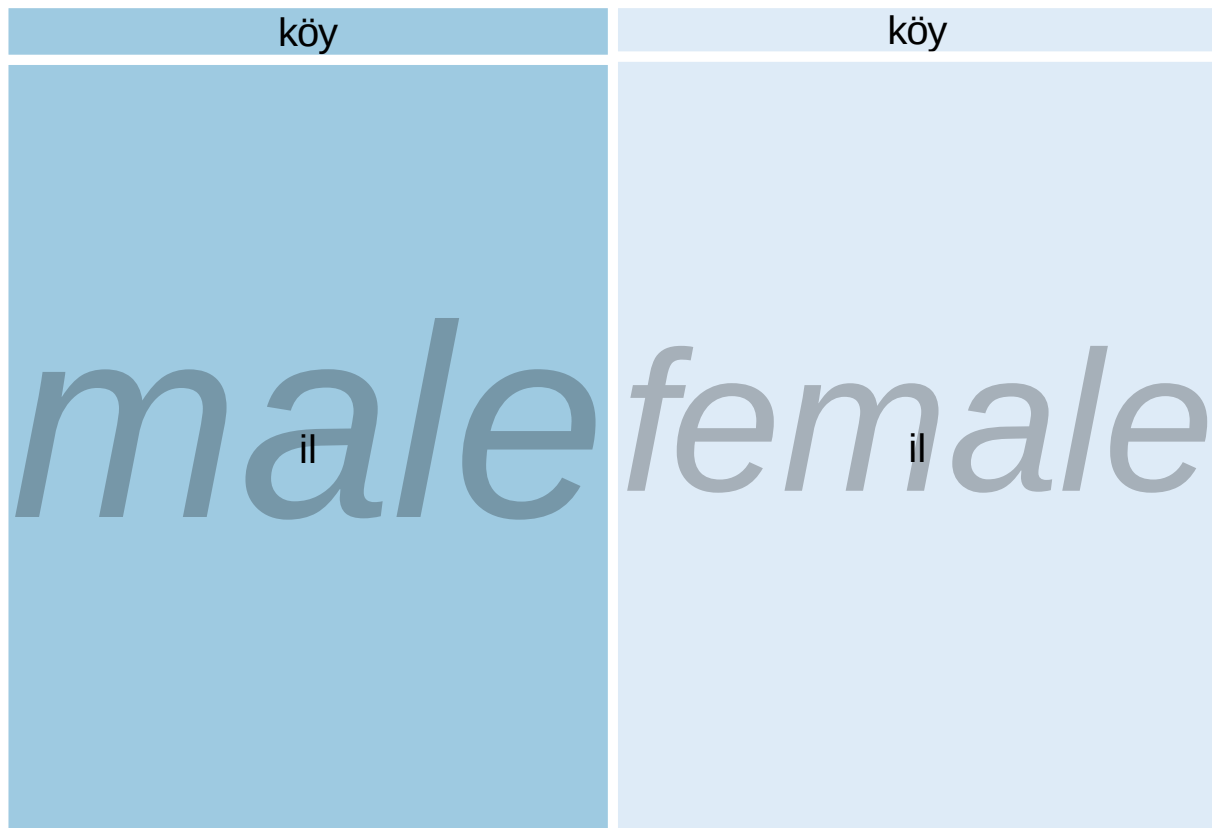
```
ggplot(poptr2, aes(area = pop2,
  fill = sex,
  label = yer,
  subgroup = sex,
  subgroup2=yer)) +
  geom_treemap() +
  geom_treemap_text(colour = "black",
    place = "centre",
    size = 15) +
  scale_fill_brewer(palette = "Blues") +
  geom_treemap_subgroup_border(colour = "white",
    size = 5) +
  geom_treemap_subgroup2_border(colour = "white",
    size = 5) +
  geom_treemap_subgroup_text(place = "centre",
    grow=TRUE,
    alpha = 0.25,
    colour = "black",
    fontface = "italic") +
  theme(legend.position = "none")
```

```
## Warning: Removed 77 rows containing missing values or values outside the scale range
## (`geom_treemap()`).

## Warning: Removed 77 rows containing missing values or values outside the scale range
## (`geom_treemap_text()`).

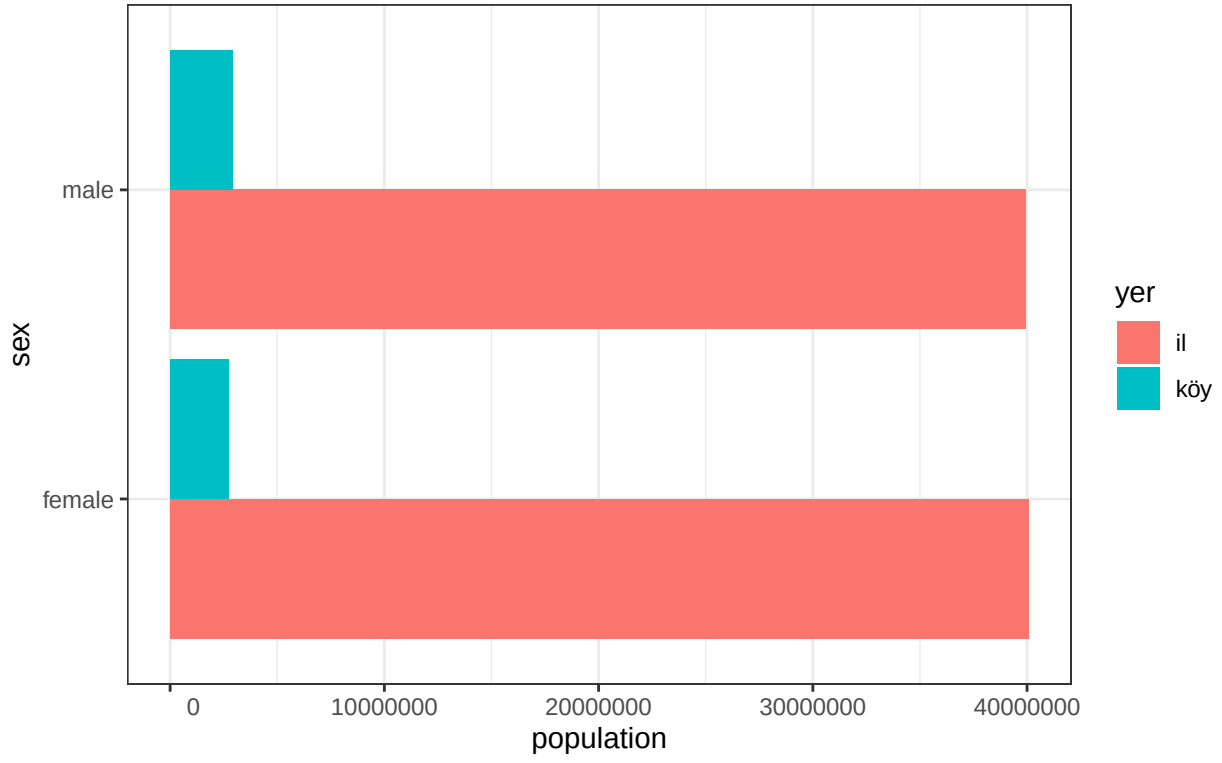
## Warning: Removed 77 rows containing missing values or values outside the scale range
## (`geom_subgroup_border()`).
## Removed 77 rows containing missing values or values outside the scale range
## (`geom_subgroup_border()`).

## Warning: Removed 77 rows containing missing values or values outside the scale range
## (`geom_subgroup_text()`).
```



```
ggplot(poptr2, aes(x = reorder(sex, +pop2), y = pop2, fill=yer )) +
  geom_bar(stat = "identity",position="dodge") +
  scale_y_continuous(labels = function(x) format(x, scientific = FALSE)) +
  labs(x      = "sex",
       y      = "population",
       title  = "yaşam alanına göre cinsiyetlerin nüfusu",
       caption = "") +
  coord_flip() +
  na.omit(poptr2)+
  theme(text = element_text(size = 20))+
  theme_bw()
```

ya.am alan.na göre cinsiyetlerin nüfusu



graph showing the population living in villages and provinces, broken down by gender, is provided. While the proportion of males and females is roughly the same in the provinces, the proportion of females is less than that of males in the villages.